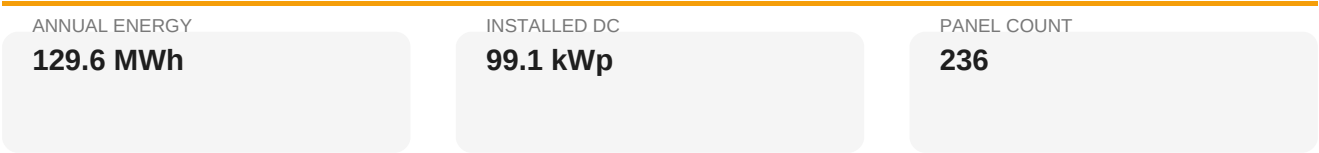
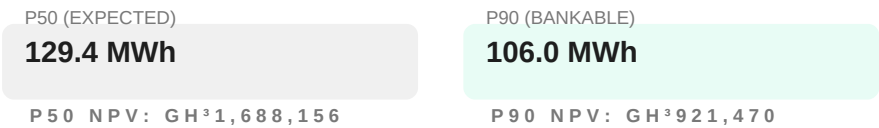


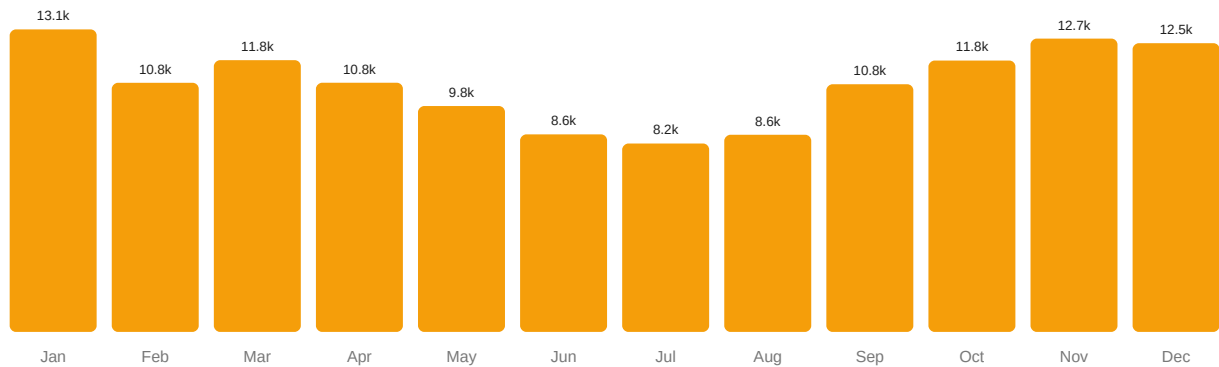
Yield Summary



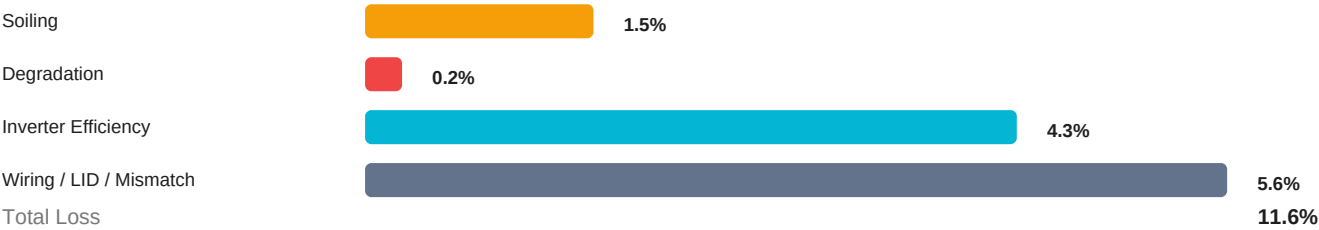
Yield Probability Distribution (1,000 Monte Carlo runs)



Monthly Yield Forecast (kWh)



Loss Waterfall



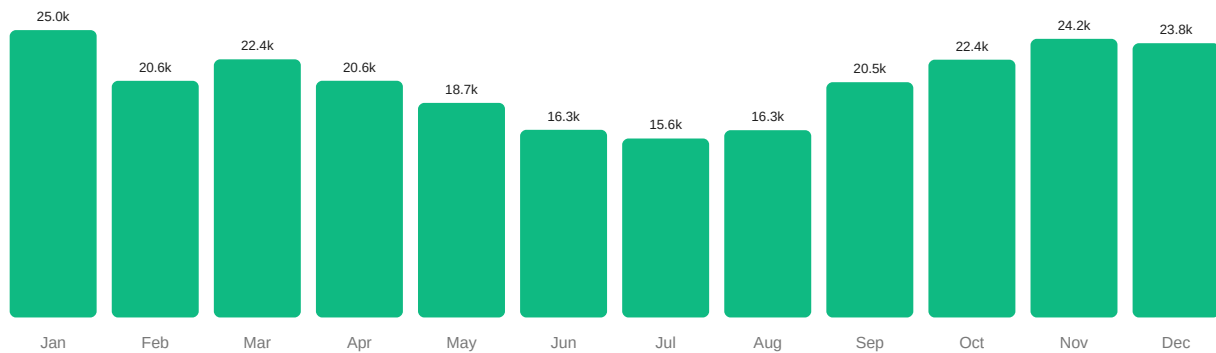
Key Assumptions

Degradation Rate	0.5%/yr	Soiling Loss	5.0% (annual avg)	Discount Rate (WACC)	8.0%
Tariff Escalation	3.0%/yr	O&M Escalation	2.0%/yr	System Lifetime	25 years
Inverter Efficiency	95.7%	Wiring Loss	2.0%	LID Loss	2.0%
Mismatch Loss	2.0%	DC/AC Ratio	~1.2 (module-dependent)		

Financial Summary

NET PRESENT VALUE G H ³ 1,694,067	PAYBACK PERIOD 6.7 Years	INTERNAL RATE OF RETURN 15.8%
LCOE G H ³ 1.83/kW h	TOTAL CAPEX G H ³ 1,982,400	LIFETIME SAVINGS G H ³ 7,501,888

Projected Monthly Savings (G H ³, tariff G H ³ 1.90/kW h)



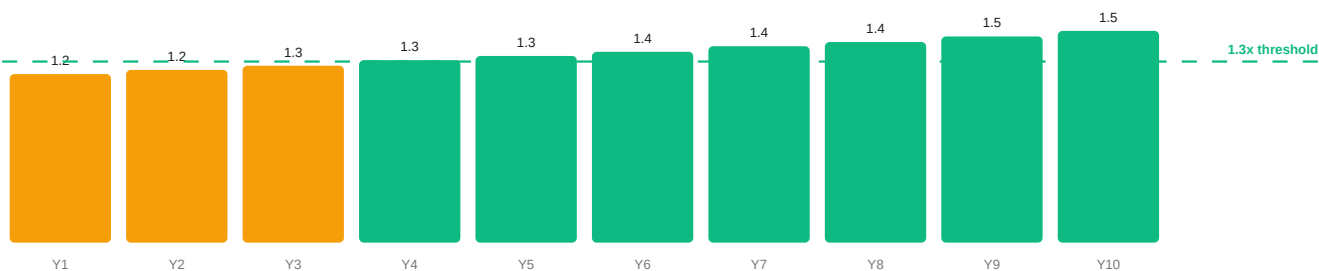
Financial Parameters

Installed DC	99.1 kWp
LCOE	G H ³ 1.83 / kW h
Year-1 Savings	G H ³ 246,286
Tariff Mode	Flat Rate

Debt Service Coverage Ratio (DSCR)

65% debt at 12% interest over 10 years

DEBT AMOUNT G H ³ 1,288,560	EQUITY G H ³ 693,840	ANNUAL DEBT SERVICE G H ³ 228,055	MIN DSCR 1.21x
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O&M Cost Breakdown

G H ³ 320/kWp/year — per IRENA West Africa benchmarks

CLEANING G H ³ 80/kW p	INVERTER RESERVE G H ³ 61/kW p	MONITORING G H ³ 38/kW p	INSURANCE G H ³ 51/kW p	SPARE PARTS G H ³ 90/kW p
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Sensitivity Analysis — NPV Tornado

±20% variation in each parameter (all else held constant)



Base NPV: GH 1,694,067

Environmental Impact

PM2.5 (Ambient)	Not yet available
PM10 (Ambient)	Not yet available
Cleaning Events / Month	6.4

PM data requires ground-level sensors; satellite sources (NASA POWER) do not provide particulate measurements.
Cleaning events estimated from rainfall events (>0.5 mm); heavy rain self-cleans panels.

Carbon Equivalence

Based on 129.6 MWh annual yield, this system avoids an estimated 64.8 tonnes CO₂,
(Using Ghana grid emission factor of ~0.5 kg CO₂/kWh)

Uncertainty Disclosure

Monte Carlo P50/P90 spread reflects combined uncertainty from inter-annual irradiance variability (±5%), Harmattan soiling deposition (±12%), hardware tolerance (±3%), tariff regulation risk (±15%), degradation variance (±0.2%/yr), and grid availability (±5%). P90 represents the yield exceeded 90% of simulated years — suitable for bankable yield estimates per IEC 61724 methodology.